

MATH 301: Homework 5

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Section 3.3 : 2 Let $x_1 > 1$ and $x_{n+1} := 2 - 1/x_n$ for $n \in \mathbb{N}$. show that (x_n) is bounded and monotone. Find the limit.

Monotone Sequence Let $X = (x_n)$ be a sequence.

- X is increasing if $x_1 \leq x_2 \leq \dots \leq x_n \leq \dots$
- X is decreasing if $x_1 \geq x_2 \geq \dots \geq x_n \geq \dots$
- X is monotone if it is either increasing or decreasing.

Note the following relationship:

$$1 < x_1 \implies \frac{1}{x_1} < 1 \quad \text{Let } \lambda_n := \frac{1}{x_n}$$

Observe first few values of X :

$$x_2 = 2 - \lambda_1 \quad x_3 = 2 - \lambda_2 \quad \dots \quad x_n = 2 - \lambda_{n-1}$$

The nature of the growth/decay of the function is dependent on the values of the λ term.

Notice (Subproblem A1):

$$\begin{aligned} &\text{Observe } x_1 \in (1, \infty) \text{ Suppose } x_k \in (1, \infty) \\ &\implies 1 < x_k \implies \frac{1}{x_k} < 1 \implies 0 < 1 - \frac{1}{x_k} \\ &\implies 1 < 2 - \frac{1}{x_k} \equiv 1 < x_{k+1} \\ &\therefore \forall n \in \mathbb{N}, \text{ by PMI } x_n \in (1, \infty) \end{aligned}$$

$$\text{Notice: } x_{k+1} - x_k := \left(2 - \frac{1}{x_k}\right) - x_k = \frac{x_k - 2x_k + 1 - x_k^2}{x_k} = -\frac{(x_k - 1)^2}{x_k}$$

Note from Subproblem A1 we can construct these three statements:

$$0 < x_n \quad 0 < (x_n - 1)^2 \quad -\frac{(x_k - 1)^2}{x_k} < 0$$

As $x_{k+1} - x_k < 0$. This implies $x_{k+1} < x_k$. Thus the sequence is monotonic and decreasing. Therefore a limit must exist.

Recall the limit of the tail of any sequence must converge to the same limit of the sequence as for any sequence we are guaranteed the uniqueness of any limit.

$$\begin{aligned} \implies \lim_{n \rightarrow \infty} x_n = L &\implies \lim_{n \rightarrow \infty} x_{n+1} = L \implies L = 2 - \frac{1}{L} \\ &\therefore L = 1 \end{aligned}$$

Section 3.3 : 3 Let $x_1 \geq 2$ and $x_{n+1} := 1 + \sqrt{x_n - 1}$ for $n \in \mathbb{N}$. Show that (x_n) is decreasing and bounded below by 2. Find the limit.

Section 3.3 : 9 Let A be an infinite subset of $\mathbb{R}$ that is bounded above and let $u := \sup A$. Show there exists an increasing sequence (x_n) with $x_n \in A$ for all $n \in \mathbb{N}$ such that $u = \lim(x_n)$.

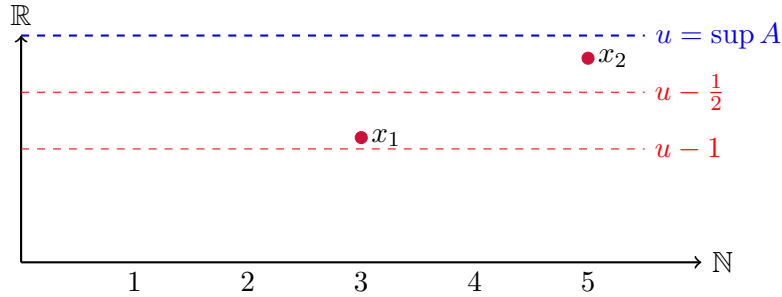
If $A \subseteq \mathbb{R}$ and $\sup A = u \implies A := \{a_1, a_2, \dots, a_n, a_{n+1}, \dots \mid a \in \mathbb{R}\}$ and $\forall a \in A, a \leq u$.

$$\sup A = u \implies \forall n \in \mathbb{N}, \exists a'_n \in A \text{ s.t. } u - \frac{1}{n} < a'_n < u$$

$$\text{Let } x_1 := u - 1 < a'_1 < u \quad \text{Let } x_2 := \max(x_1, a'_2)$$

$$\text{Generally, } x_n := \max(x_{n-1}, a'_n)$$

Visualization



We are defining a line $u - \frac{1}{n}$ and returning the smallest value of a that is larger than $u - \frac{1}{n}$ and assigning it to the corresponding n . This a is what we have called a'_n (The minimum value of a which depends on n that shows that $u - \epsilon$ is not an upper bound)

Consider we have constructed in a sequence defined to be : $x_{n-1} \leq x_n$. Thus the sequence is increasing.

Notice as $x_n := \max(x_{n-1}, a'_n)$. This implies $a'_n \leq x_n$. Thus $u - \frac{1}{n} < a'_n \implies u - \frac{1}{n} < x_n$

Therefore for all terms $u - \frac{1}{n} < x_n \leq u$

$$\begin{aligned} &\lim_{n \rightarrow \infty} \left(u - \frac{1}{n} < x_n \leq u \right) \\ &u \leq x_n \leq u \implies \lim_{n \rightarrow \infty} x_n = u \end{aligned}$$

Section 3.3 : 11 Let $x_n := 1/1^2 + 1/2^2 + \cdots + 1/n^2$ for each $n \in \mathbb{N}$. Prove that (x_n) is increasing and bounded, and hence converges. [*Hint*: Note that if $k \geq 2$, then $1/k^2 \leq 1/k(k-1) = 1/(k-1) - 1/k$.]